May 8, 2026

Editorial Board
Annals of Henri Poincaré
Birkhäuser/Springer

Dear Editors,

I submit the manuscript *Geometric Origin of CP Phases from Hyperbolic Holonomy* for consideration in the *Annals of Henri Poincaré*.

The paper shows that CP-violating phases in the Standard Model arise naturally as holonomy invariants of flat $U(1)$ connections on compact hyperbolic 3-manifolds. The central results are:

1. **CP obstruction theorem**: If $M = H^3/\Gamma$ admits an orientation-reversing isometry Θ, then Θ induces complex conjugation on $U(1)$ character representations of $\pi_1(M)$, forcing the Jarlskog invariant $J = 0$ identically. Chirality of M (absence of such Θ) is necessary for $J \neq 0$.

2. **Topological stability**: For flat connections, CP phases are topological invariants stable under all metric deformations preserving topological type.

3. **Explicit realization**: The Meyerhoff manifold (OrientableClosedCensus[1], $H_1 = Z/5$, vol $= 0.9814$) is chiral and realizes $J \neq 0$, while its CKM counterpart (OrientableClosedCensus[43 $H_1 = Z/5$) is amphicheiral with $J = 0$, geometrically explaining the observed quark–lepton CP asymmetry.

The paper is purely mathematical in character—the argument is formulated entirely in representation-theoretic language—with a physical interpretation in terms of Standard Model observables. It sits squarely at the intersection of hyperbolic geometry, topological field theory, and mathematical physics that AHP addresses.

The manuscript has not been submitted elsewhere and is not under consideration by another journal.

Sincerely,

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